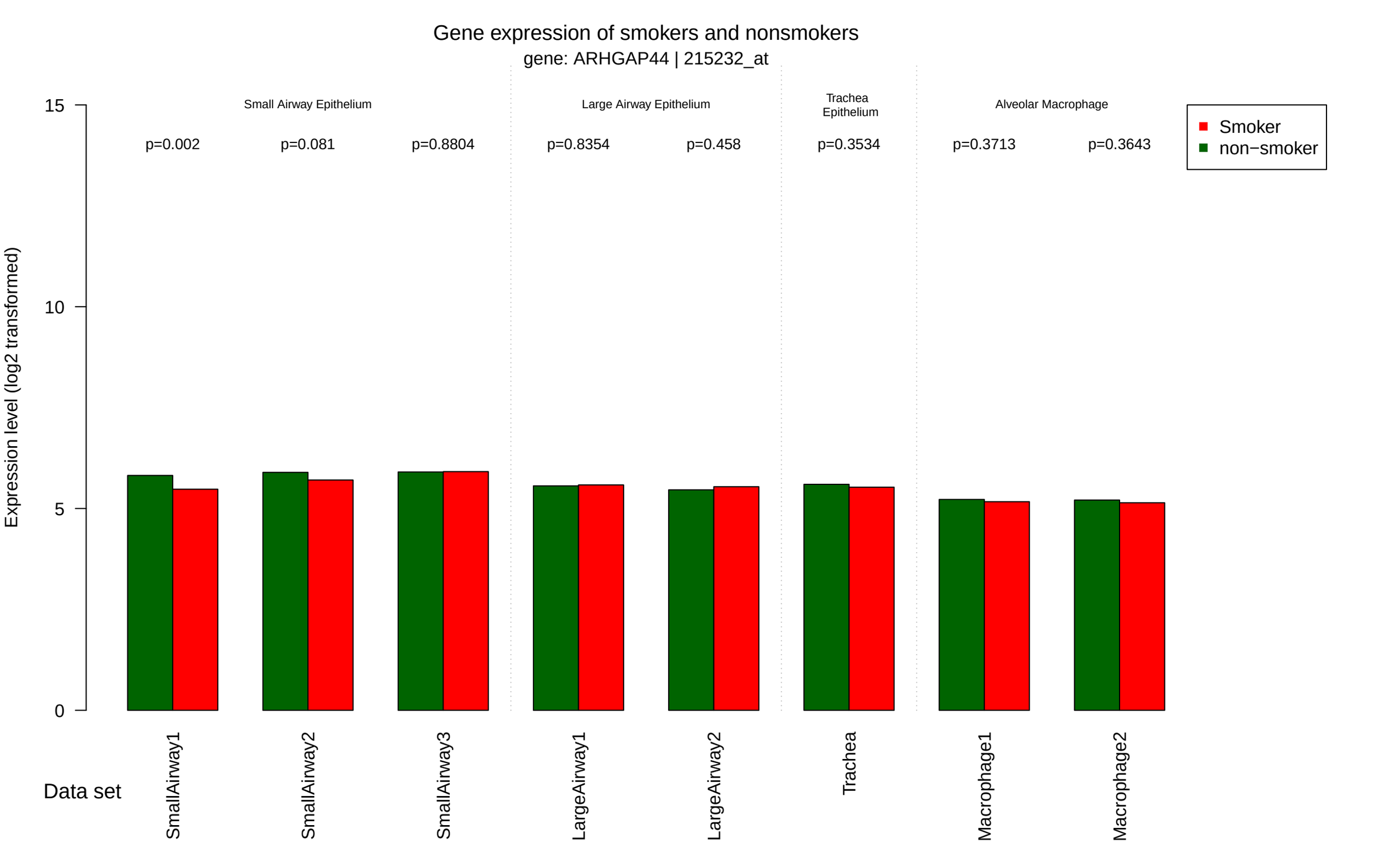
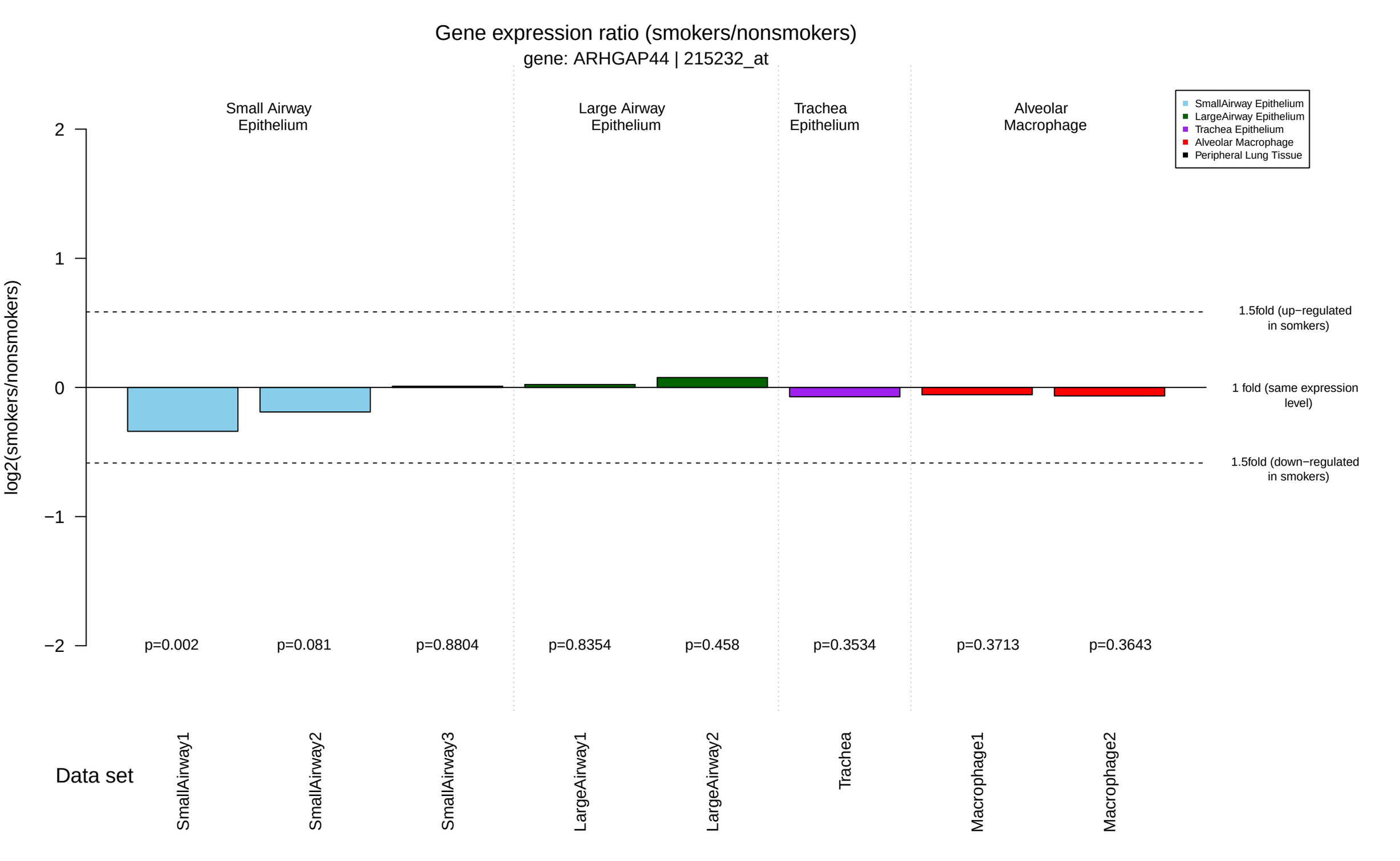
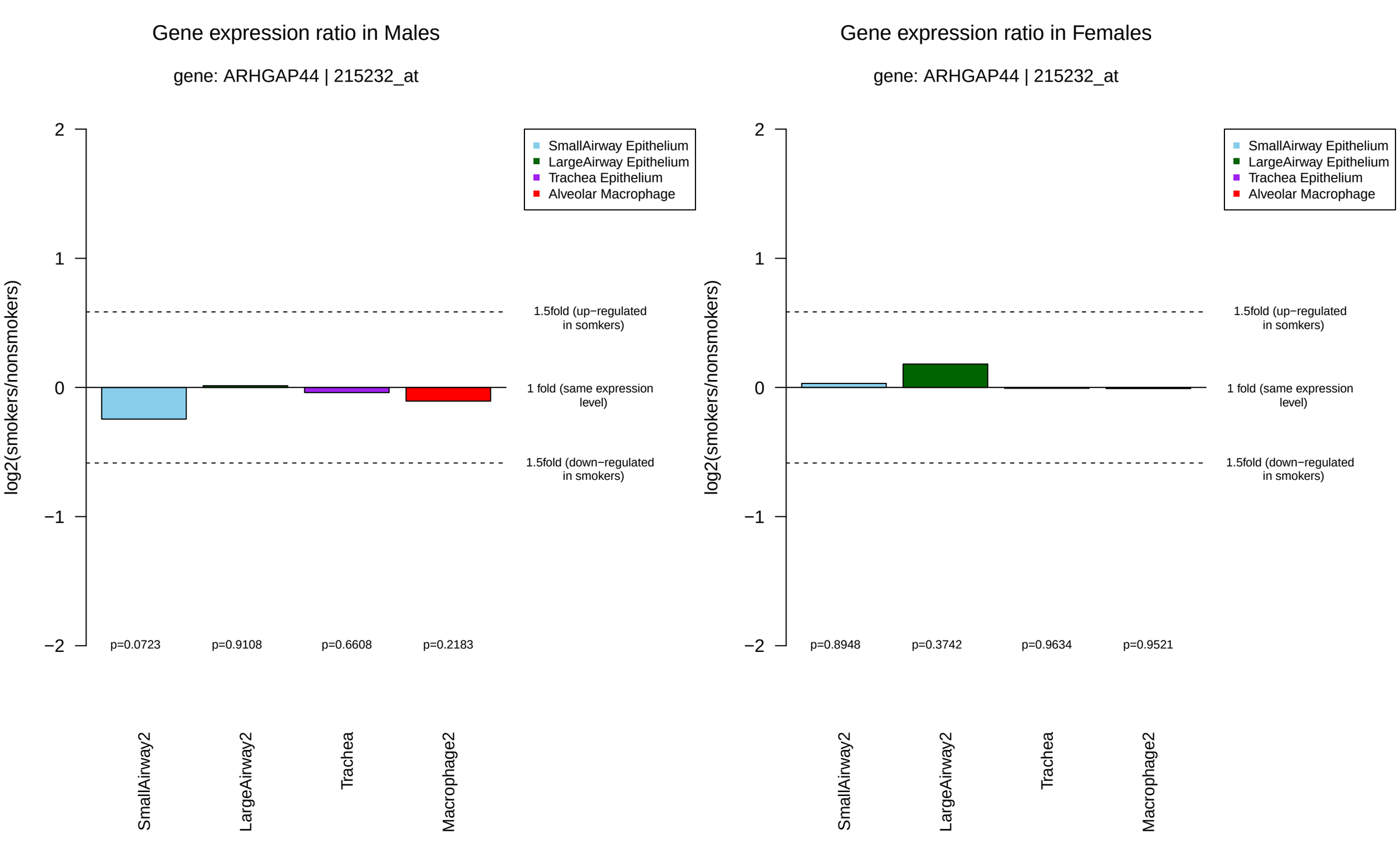
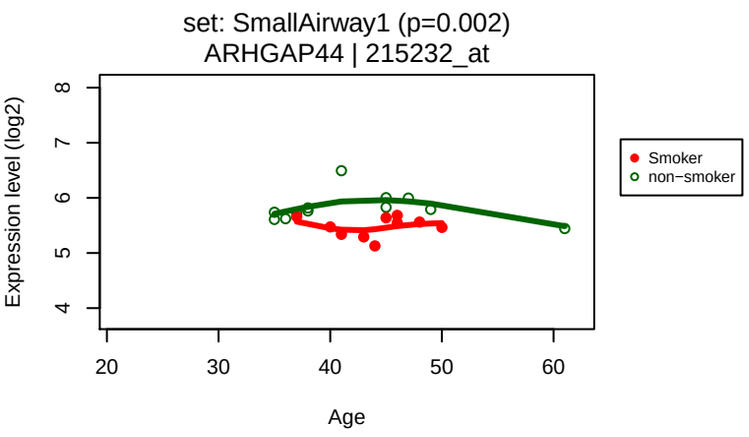
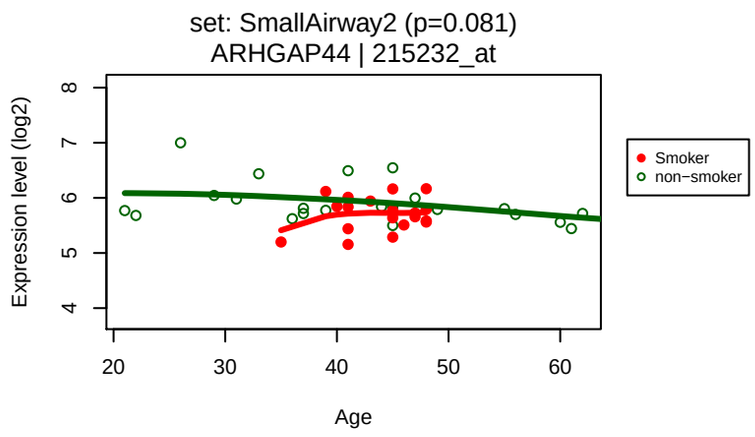
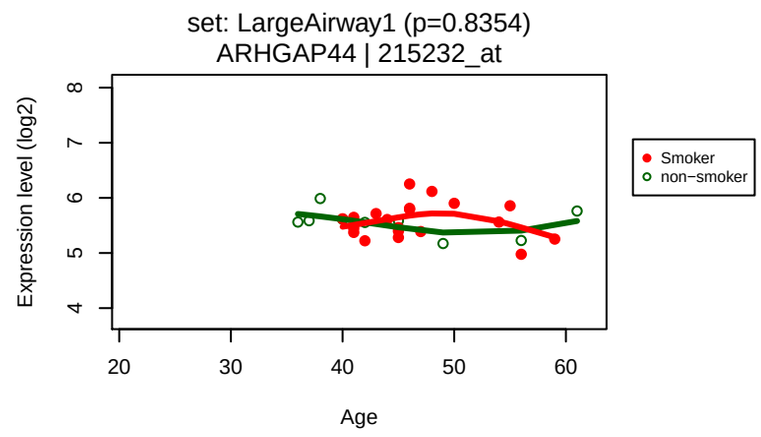

Gene expression of smokers and nonsmokers
gene: ARHGAP44 | 215232_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.002
p=0.081
p=0.8804
p=0.8354
p=0.458
p=0.3534
p=0.3713
p=0.3643
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

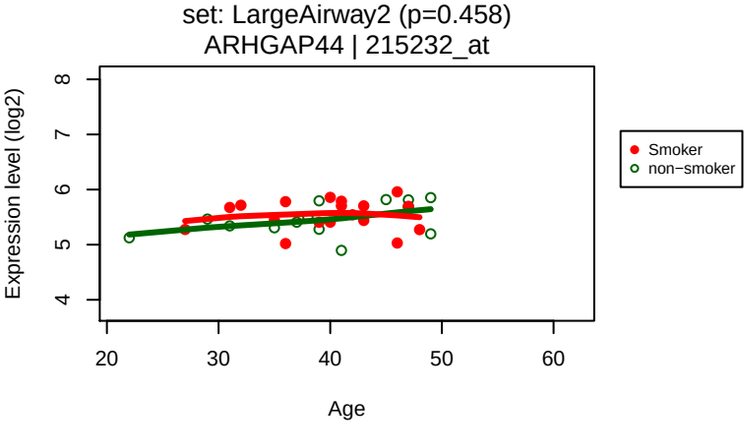

Gene expression ratio (smokers/nonsmokers)
gene: ARHGAP44 | 215232_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.002
p=0.081
p=0.8804
p=0.8354
p=0.458
p=0.3534
p=0.3713
p=0.3643
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

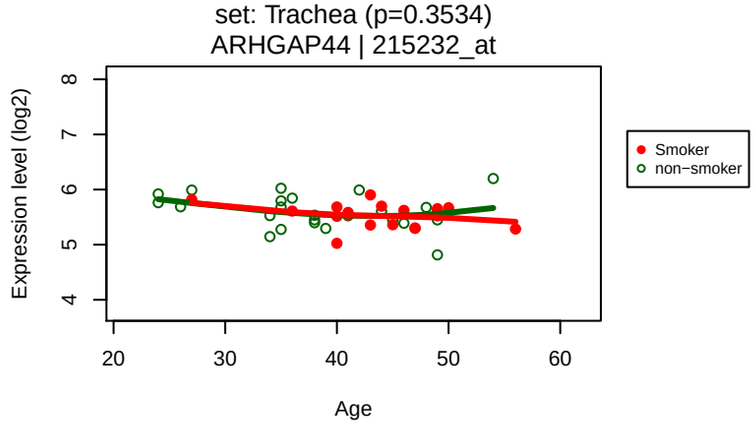

Gene expression ratio in Males
gene: ARHGAP44 | 215232_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0723
p=0.9108
p=0.6608
p=0.2183
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: ARHGAP44 | 215232_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.8948
p=0.3742
p=0.9634
p=0.9521
SmallAirway2
LargeAirway2
Trachea
Macrophage2

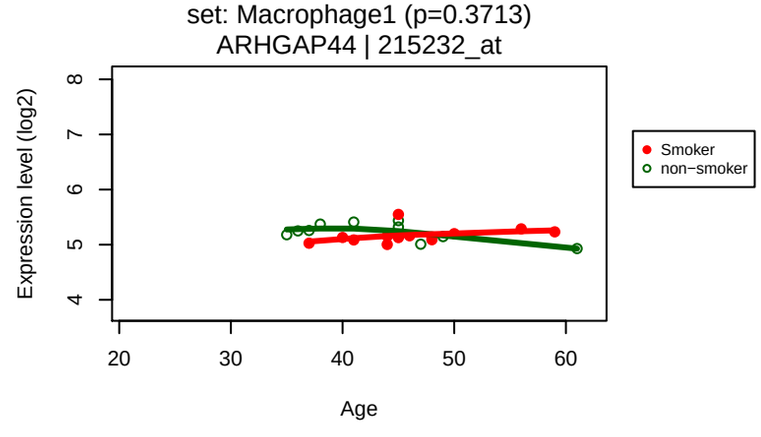

set: SmallAirway1 (p=0.002)
ARHGAP44 | 215232_at
Expression level (log2)
Age
Smoker
non−smoker

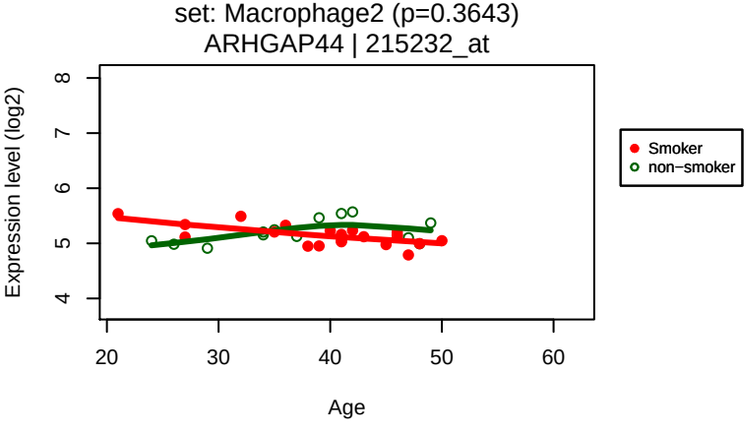

set: SmallAirway2 (p=0.081)
ARHGAP44 | 215232_at
Expression level (log2)
Age
Smoker
non−smoker

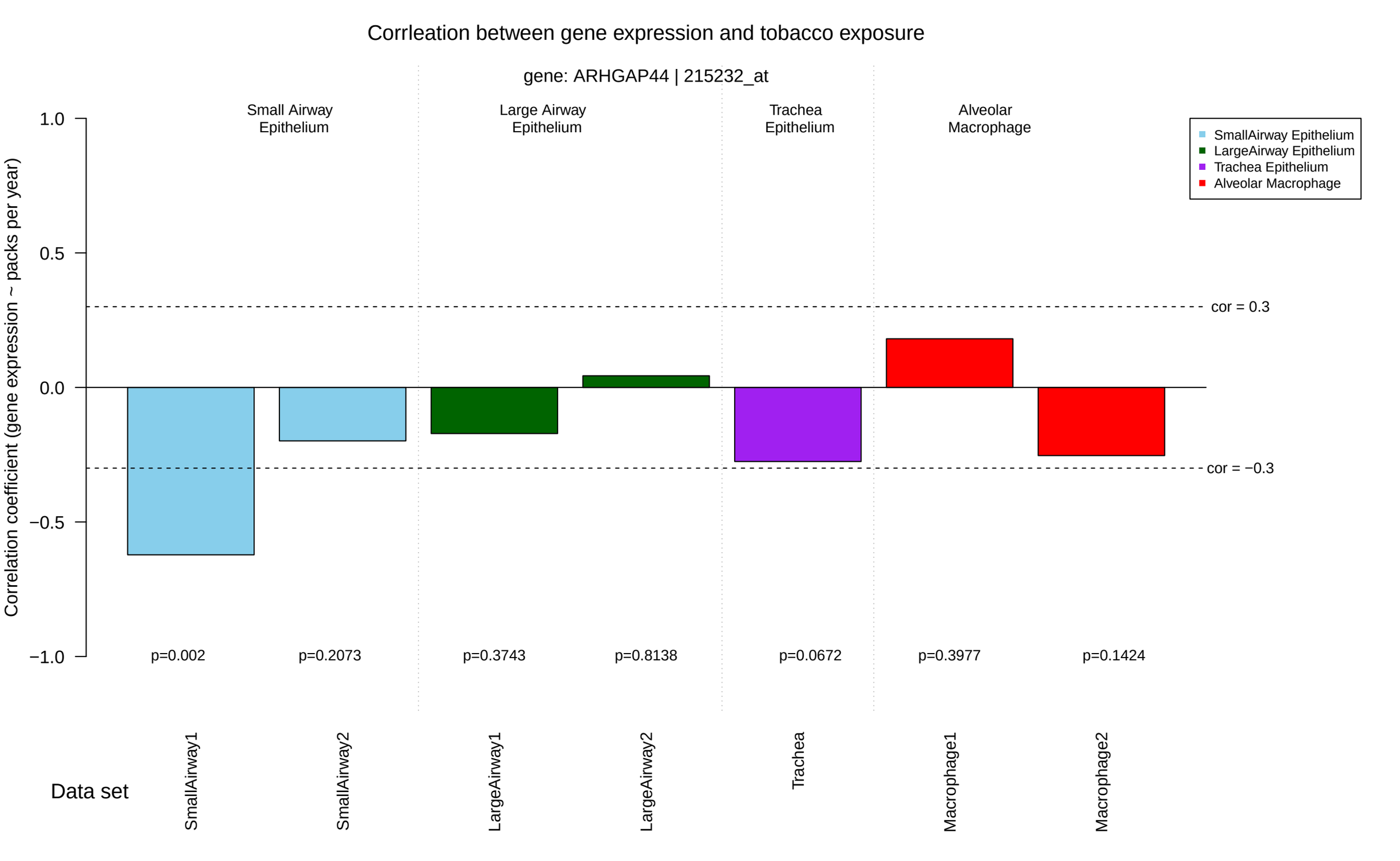

set: LargeAirway1 (p=0.8354)
ARHGAP44 | 215232_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.458)
ARHGAP44 | 215232_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.3534)
ARHGAP44 | 215232_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.3713)
ARHGAP44 | 215232_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.3643)
ARHGAP44 | 215232_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: ARHGAP44 | 215232_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.002
p=0.2073
p=0.3743
p=0.8138
p=0.0672
p=0.3977
p=0.1424
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2